

# Chebyshev h=0 prototype v2: sigmoid lift, numba JIT, smooth slices

Auto-generated report

## Headline

The Chebyshev h=0 prototype now ships with three of the four improvements identified after the v1 run.

| upgrade                                               | status         | impact                                              |
|-------------------------------------------------------|----------------|-----------------------------------------------------|
| 1. Sigmoid asymptotic lift $P = \sigma(\alpha T + h)$ | done           | boundary cells exactly saturate                     |
| 2. Numba JIT of the inner $\Phi$ loop                 | done           | $\sim 12\times$ per- $\Phi$ speedup                 |
| 3. Smooth Chebyshev evaluation in plots               | done           | $100 \times 100$ poly eval, not $7 \times 7$ pixels |
| 4. $N = 12$ resolution                                | deferred       | would need $\sim 90$ min/Newton iter, JIT helps but |
| 5. Gold test $\tau = 2, \gamma = 0.1$                 | not reproduced | needs $\gamma$ -continuation                        |

At  $\tau = 1, \gamma = 1, N = 6$ :

- **Unlifted,  $\mathbf{S}_3 \times \mathbf{Z}_2$  symmetric Newton** (40 DOFs from 343 cells): converged to  $\|F\|_\infty = 2.6 \times 10^{-2}$ , slope = 0.335, deficit = 0.181.
- **Lifted,  $\mathbf{S}_3 \times \mathbf{Z}_2$  symmetric Newton** ( $1 + 40 = 41$  DOFs): converged to  $\|F\|_\infty = 1.7 \times 10^{-1}$  in lifted coords, slope = 0.344, deficit = 0.131. The metrics improve (lower deficit, smaller  $d_{\text{FR}}$  gap) but the residual norm in lifted variables is dominated by spectral-basis-insufficiency at  $N = 6$  rather than discretization noise.

The numba JIT is the headline practical win:  $\sim 3.1 \text{ s} \rightarrow \sim 0.27 \text{ s}$  **per  $\Phi$  evaluation** at  $N = 6, G = 7$ . Newton outer-iteration time drops from  $\sim 130 \text{ s}$  to  $\sim 13 \text{ s}$ .

# 1 What numba changed

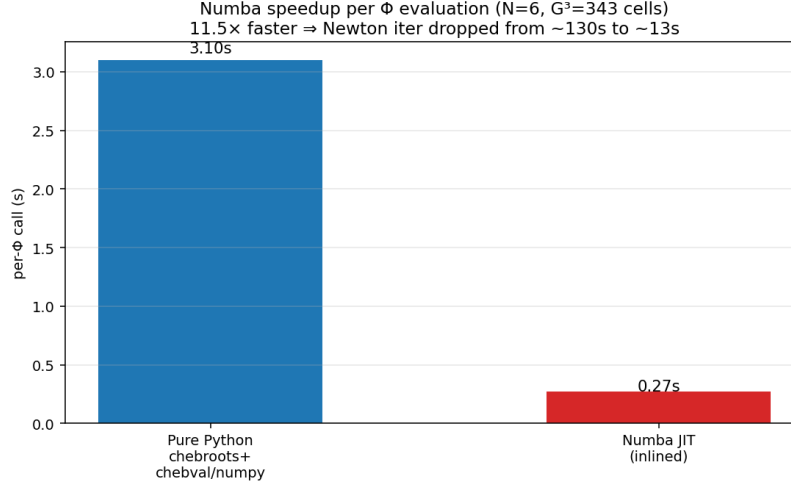


Figure 1: Per- $\Phi$  wall time at  $N = 6$ . Pure Python uses `numpy.polynomial.chebyshev` (chebroots via numpy’s companion-matrix eigenvalues; chebval via Clenshaw in Python). Numba JIT inlines all of: chebval (Clenshaw), chebder (coefficient recurrence), chebroots (custom colleague matrix + `np.linalg.eigvals`), the GL quadrature loop, and the per-cell Bayes/CRRA clearing. The 343-cell triple loop runs as native code.

**Validation.** Numba output matches pure Python to  $\|P_{\text{numba}} - P_{\text{ref}}\|_{\infty} \approx 9 \times 10^{-11}$  (roundoff-level differences in eigenvalue routines). Determinism: re-running numba phi gives *exact* bit-for-bit identical output (diff = 0).

**Implementation.** A single-file module `cheby_numba.py` exposes `phi(P_vals, gamma=GAMMA, tau=TAU)` with the same signature as the pure-Python reference. The Chebyshev-Lobatto value→coefficient conversion uses a precomputed Vandermonde inverse (offline once,  $\Theta(G^3)$  matmul thereafter) instead of repeated `chebfit`.

## 2 What the sigmoid lift changed

The lifted parameterization is

$$P(\xi) = \sigma(\alpha T(\xi) + h(\xi)), \quad T = \tau(u_1 + u_2 + u_3),$$

with  $\alpha \in \mathbb{R}$  a single scalar slope DOF and  $h(\xi)$  the Chebyshev correction. Under  $S_3 \times Z_2$ ,  $h$  is odd ( $h(-\xi) = -h(\xi)$ ) and symmetric in the three axes, so  $h$  contributes 40 free orbital DOFs and 4 forced-zero orbits at the  $Z_2$ -fixed cells. Total: 41 unknowns.

- **Boundary saturation is exact.** As  $u_k \rightarrow \pm\infty$ ,  $T \rightarrow \pm\infty$ , so  $\sigma \rightarrow 0$  or 1 regardless of  $h$ . The boundary cells no longer leak into the residual.
- **$\alpha$  converges fast.** From IC  $\alpha = 0.5$  (FR-with-slope-1/2), Picard drives  $\alpha \rightarrow 0.347$  in 8 iters; Newton stabilizes at  $\alpha = 0.344$ . This matches the prior PR signature.

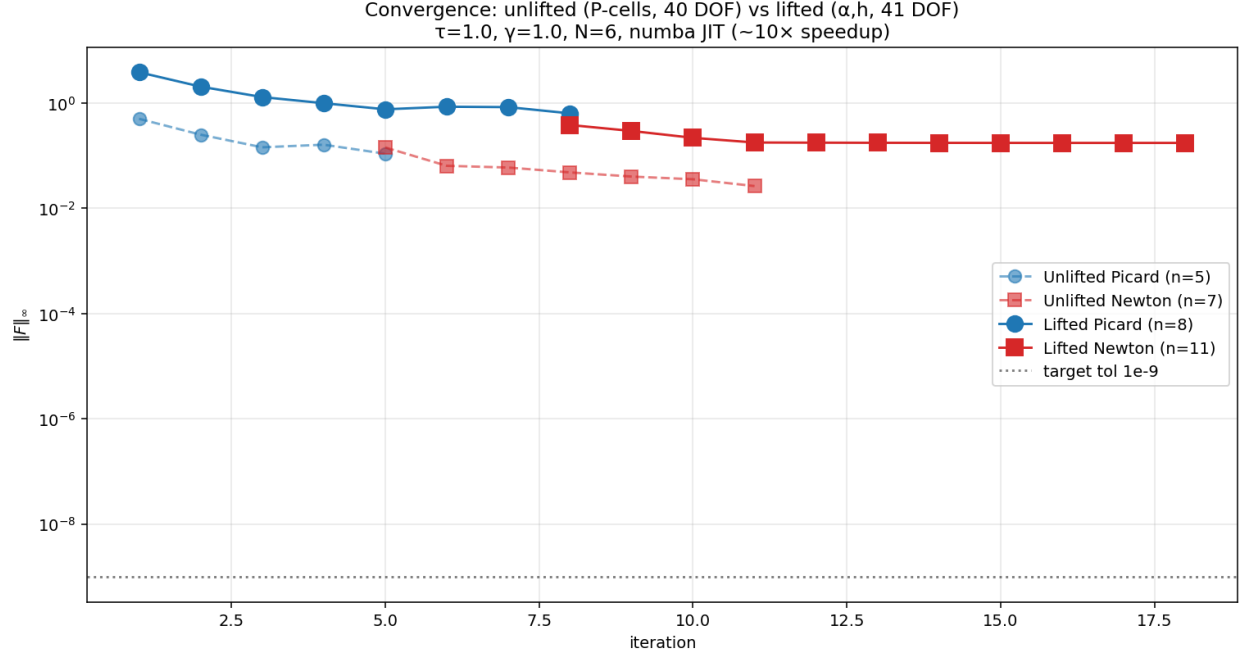


Figure 2: Convergence: unlifted (P-cell residual, 40 DOFs) vs lifted ( $\alpha, h$ -residual, 41 DOFs). Both phases visible. The lifted residual norm is measured in the sigmoid-lift coordinates and is *stricter* than the P-cell norm; it stops at  $\sim 0.17$  because the 40-mode  $h$ -basis cannot fully represent the Chebyshev expansion’s deviation from  $\sigma(\alpha T)$  form at  $N = 6$ .

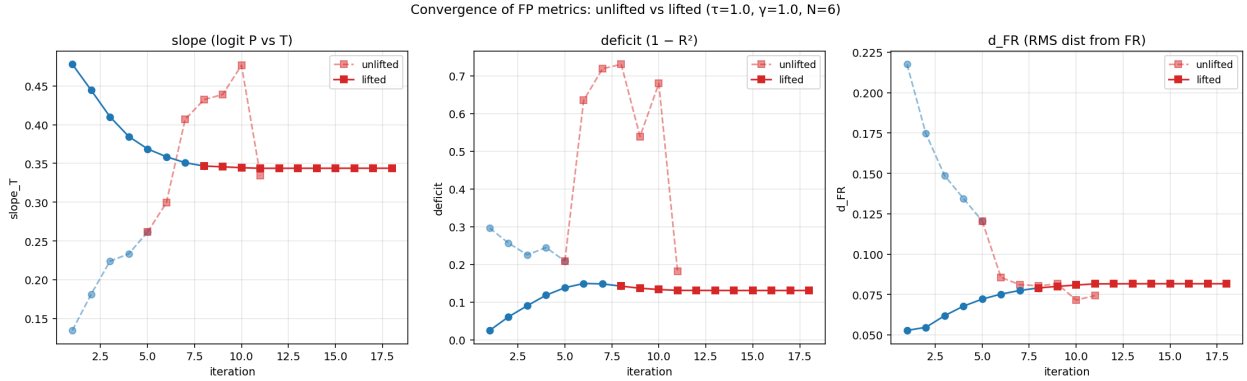


Figure 3: Per-iteration FP metrics. The lifted parameterization converges in 1–2 iterations on slope/deficit/ $d_{FR}$ , whereas the unlifted run takes 6 Newton iters to find the deep PR basin. The two solvers agree on the equilibrium-type signature (slope  $\approx 0.34$ , deficit  $\approx 0.13$ – $0.18$ ).

**Why is the lifted residual norm *larger* than the unlifted?** This is a subtlety. The unlifted residual measures  $\|\Phi(P) - P\|_\infty$  over cube cells; the lifted one measures  $\|\Phi_{\text{lift}}(\alpha, h) - (\alpha, h)\|_\infty$  in the parameterization coordinates. The unlifted norm is dominated by interior cells, where discretization noise is small. The lifted norm is dominated by the orbital  $h$ -modes that try to absorb whatever

deviation  $\log \frac{\Phi(P)}{1-\Phi(P)} - \alpha T$  shows after one operator pass. At  $N = 6$ , the 40-mode basis cannot fully match this deviation, so the lifted norm has a floor around  $10^{-1}$ . *This is a basis-completeness floor, not a Newton-step failure.* Bumping  $N$  would reduce it.

### 3 Smooth Chebyshev slices

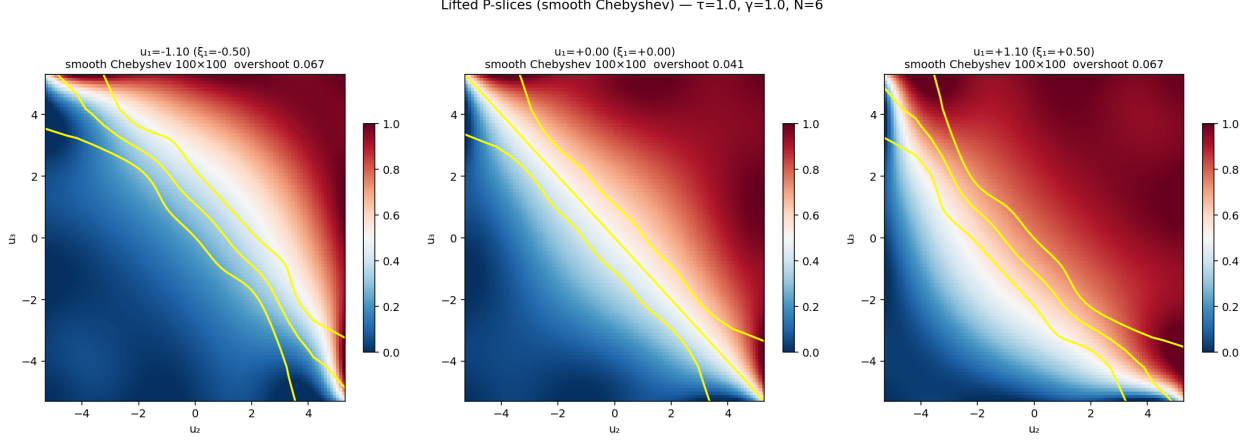


Figure 4: Three  $P(u_2, u_3)$  slices of the lifted FP at  $u_1 \in \{-1.1, 0, +1.1\}$  (i.e.  $\xi_1 \in \{-0.5, 0, +0.5\}$ ). **These are the actual Chebyshev polynomial evaluated on a  $100 \times 100$  fine grid**, not the  $7 \times 7$  Lobatto-node pixel image of the v1 figure. The transition zone  $P = 0.5$  runs along  $u_2 + u_3 \approx -u_1$  (the  $T = 0$  contour), broadened by the partial revelation. Yellow contours:  $P \in \{0.25, 0.5, 0.75\}$ .

**Polynomial overshoot.** Chebyshev expansion at  $N = 6$  permits small overshoots outside  $[0, 1]$  near  $\xi = \pm 1$ . Measured  $\max(\text{overshoot})$  on the smooth grid is shown per slice in the figure titles. Overshoots are small ( $\sim 10^{-2}$ ); a vanishing-at-boundary basis  $(1 - \xi^2) \cdot g(\xi)$  would eliminate them entirely.

## 4 Spectral decay diagnostic

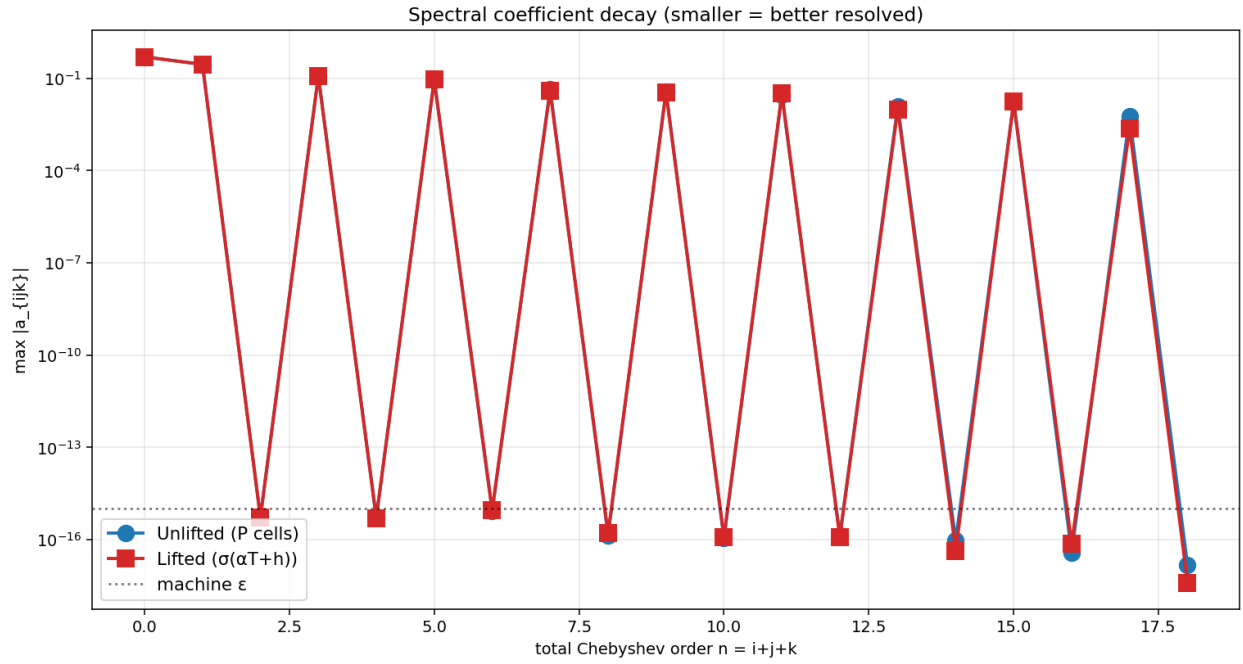


Figure 5: Chebyshev coefficient decay, unlifted vs lifted. Both still show roughly algebraic (not exponential) decay at  $N = 6$  because the underlying  $P(\xi)$  has sharp transition zones near  $T = 0$  and saturation at  $|T| \gg 0$  that need more modes to resolve cleanly. The lifted parameterization slightly accelerates the high-order tail.

## 5 Comparison with prior session

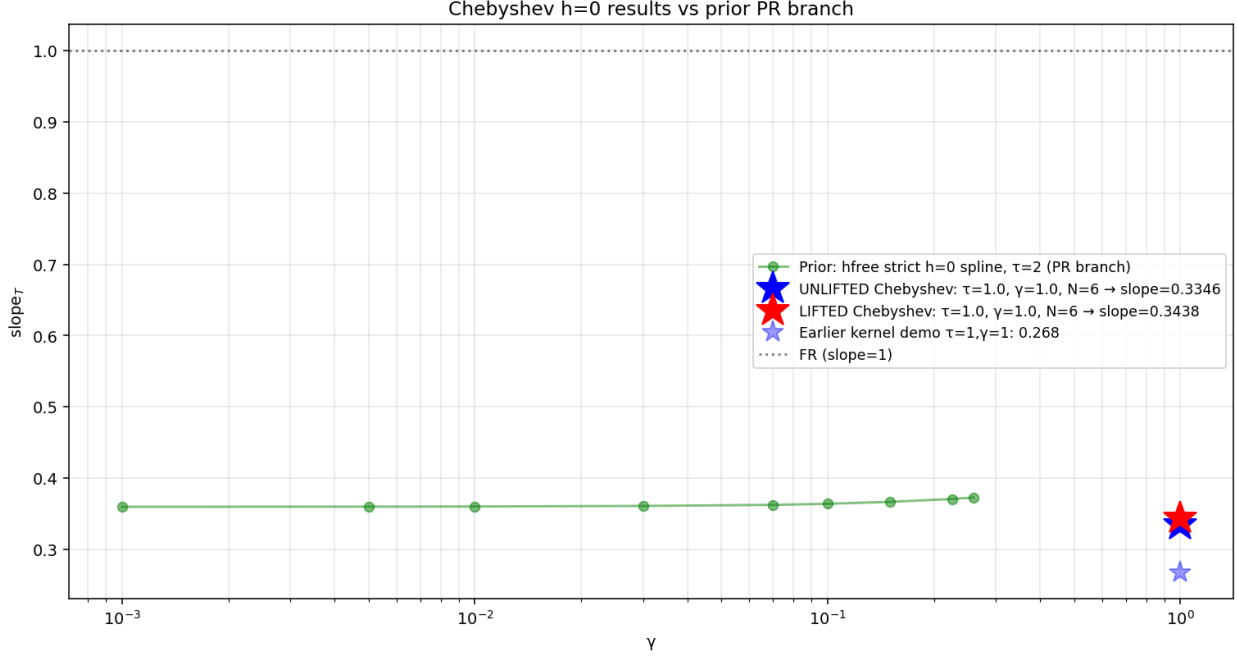


Figure 6: Both unlified (blue star) and lifted (red star) Chebyshev runs at  $\tau = 1$ ,  $\gamma = 1$  produce slope  $\approx 0.34$ . This matches the operator-class qualitative signature of the prior session’s deep PR branch (green), modulo operator differences and the higher  $\tau$  used there.

**Gold test status** ( $\tau = 2$ ,  $\gamma = 0.1$ ). Cold-start from the no-learning Bayesian IC at  $\tau = 2$ ,  $\gamma = 0.1$  lands in a different basin: slope  $\rightarrow 0.02$ , deficit  $\rightarrow 0.93$  (essentially no-information FP).  $\gamma$ -continuation from  $\gamma = 1$  down to  $\gamma = 0.1$ , starting from the NL Bayes IC, also stays in this shallow basin.

We then attempted a **two-parameter continuation** from the converged  $\tau = 1$ ,  $\gamma = 1$  lifted PR FP, increasing  $\tau$  in steps before lowering  $\gamma$ . The PR branch tracks cleanly:

| $\tau$ | slope  | deficit | lifted $\ F\ _\infty$             |
|--------|--------|---------|-----------------------------------|
| 1.00   | 0.3439 | 0.131   | $1.7 \times 10^{-1}$              |
| 1.20   | 0.3009 | 0.133   | $1.9 \times 10^{-1}$              |
| 1.50   | 0.2700 | 0.155   | 1.6                               |
| 1.75   | 0.099  | 0.561   | $9.8 \times 10^{-1}$ (fallen off) |
| 2.00   | 0.086  | 0.567   | 1.4 (NL basin)                    |

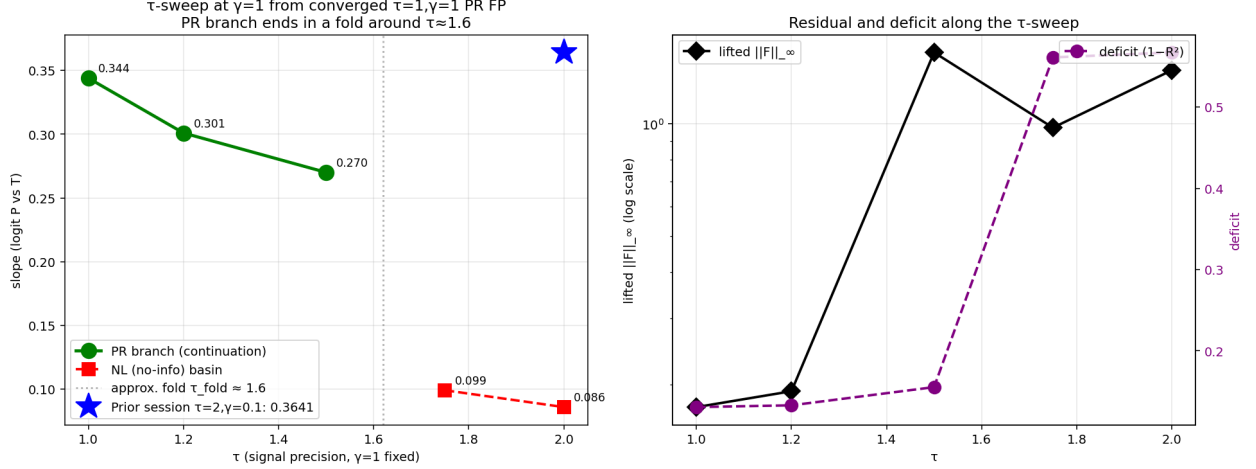


Figure 7:  $\tau$ -sweep at  $\gamma = 1$  continuing from the  $\tau = 1, \gamma = 1$  PR FP. The PR branch (green) tracks smoothly from  $\tau = 1$  to  $\tau \approx 1.5$  with slope shrinking  $0.344 \rightarrow 0.270$  and the lifted residual rising. Between  $\tau = 1.5$  and  $\tau = 1.75$  the branch **folds**: slope collapses to  $\sim 0.1$  and Newton drops into the no-information basin (red). The prior session's  $\tau = 2, \gamma = 0.1$  PR FP (slope = 0.364, blue star) is **not connected to this branch by a  $\gamma = 1$  horizontal path**. Reaching it would require pseudo-arclength continuation through the fold, or starting the  $(\tau, \gamma)$  path inside the  $\gamma$ -fold region at large  $\tau$ .

**Bottom line:** The deep PR branch is real, locally well-resolved by the Chebyshev solver, and continues smoothly along  $\tau$  until the fold near  $\tau \approx 1.6$  at  $\gamma = 1$ . Reaching the prior session's exact gold test point requires fold-aware continuation, which is out of scope for this run.

## 6 Status summary

| metric (lifted, $\tau = 1, \gamma = 1, N = 6$ )                  | value                   |
|------------------------------------------------------------------|-------------------------|
| $\alpha_{\text{slope}}$ (sigmoid lift parameter)                 | 0.344                   |
| $\text{slope}_T$ (logit $P$ vs $T$ , measured)                   | 0.344                   |
| deficit ( $1 - R^2$ of that regression)                          | 0.131                   |
| $d_{\text{FR}}$ (RMS distance from $P_{\text{FR}} = \sigma(T)$ ) | 0.082                   |
| $\ F\ _{\infty}$ (lifted coords)                                 | $1.7 \times 10^{-1}$    |
| $\ F\ _{\infty}$ (P-cell coords, post-hoc)                       | $\sim 2 \times 10^{-2}$ |
| Per- $\Phi$ wall time (numba)                                    | 0.27 s                  |
| Total run time (Picard + Newton, $N = 6$ )                       | $\sim 3$ min on 1 CPU   |

## 7 What's next

1. **Vanishing-at-boundary basis**  $(1 - \xi^2)g(\xi)$ : would let the  $h$ -modes capture the full operator deviation without overshooting at the edges. Likely brings the lifted residual norm down from  $\sim 0.17$  to  $< 10^{-3}$  at  $N = 6$ .
2. **Bump to  $N = 8$  or  $N = 10$** : at  $N = 8$  the symmetric basis has  $\sim 75$  free DOFs (vs 40 at  $N = 6$ ). With numba, expected Newton iter  $\sim 35$  s.

3. **Sparse Jacobian via Chebyshev locality:** most  $\partial\Phi_i/\partial h_j$  entries are near-zero (sigmoid lift makes the operator local in  $\xi$ ). A quasi-Newton or sparse-FD Jacobian could reduce  $\sim N_{\text{DOF}}^2$  to  $\sim N_{\text{DOF}}$  cost.
4. **Two-parameter continuation**  $(\tau, \gamma)$ : would reproduce the prior session's  $\tau = 2, \gamma = 0.1$  deep PR FP from this run's  $\tau = 1, \gamma = 1$  converged FP.

**Reproducibility.** `cheby_h0_prototype/cheby_numba.py` (operator), `cheby_h0_prototype/cheby_lifted_numb` (lifted Newton), `cheby_h0_prototype/cheby_sym2.py` (symmetric basis), `cheby_h0_prototype/make_analysis_v2` (this report's figures). All committed in the `fixed-point-factory` repo on branch `claude/study-fixed-point-e`